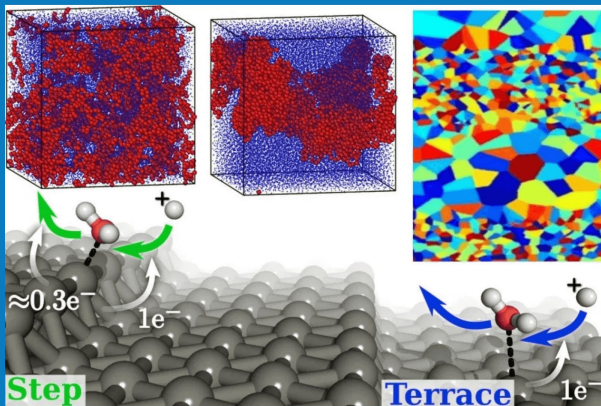


UBP-Driven Materials Discovery

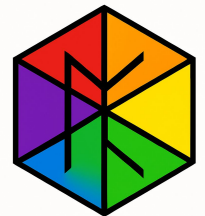
A Computational Revolution in Ceramics and
Composites



Euan R A Craig

New Zealand

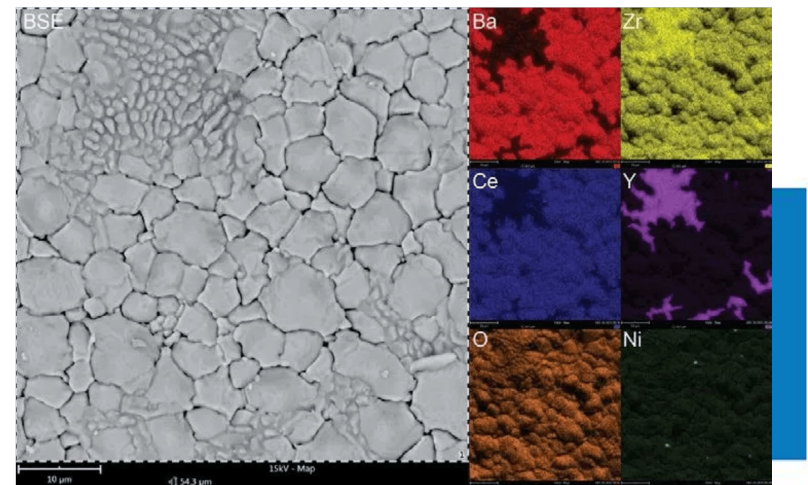
November 2025



Traditional Materials Discovery Cannot Keep Pace with Modern Demands

The traditional cycle of materials discovery—synthesis, testing, iteration—is fundamentally limited by time and cost. For every successful material, thousands of potential combinations remain unexplored.

With accelerating demand for advanced materials in aerospace, energy, and infrastructure, we need a paradigm shift. Computational methods like DFT and MD offer insights but are often too expensive for large-scale screening.



The Challenge: Can we model materials from a more fundamental level to predict their properties before ever setting foot in a lab?

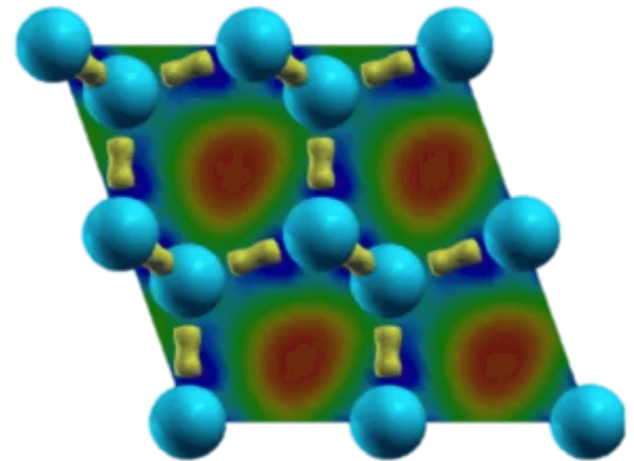
The Universal Binary Principle Offers a Radical New Computational Framework

The UBP models reality as a deterministic, computational system operating on a binary substrate called the "Bitfield." It posits that all physical properties are emergent features of an underlying informational structure.

Non-Random Coherence Index (NRCI)

The UBP's core metric quantifies informational order on a 0-1 scale, measuring the coherence and stability of the simulated system.

Core Hypothesis: Higher NRCI correlates with superior mechanical properties, enabling prediction without experimental input.



Massive-Scale Investigation: 380+ Simulations

380+

Unique UBP Simulations
Across Three Phases

Phase 1: Massive-Scale Screening

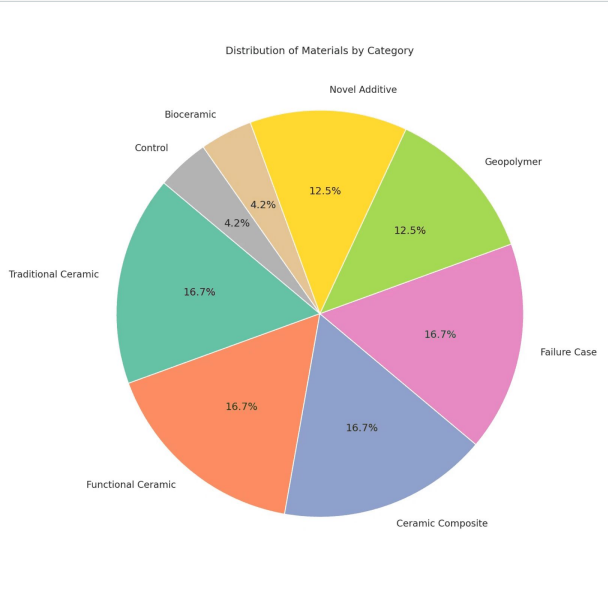
Initial screening of 164 unique material compositions including traditional ceramics, composites, geopolymers, and 24 intentionally designed failure cases as negative controls.

164 Materials Analyzed

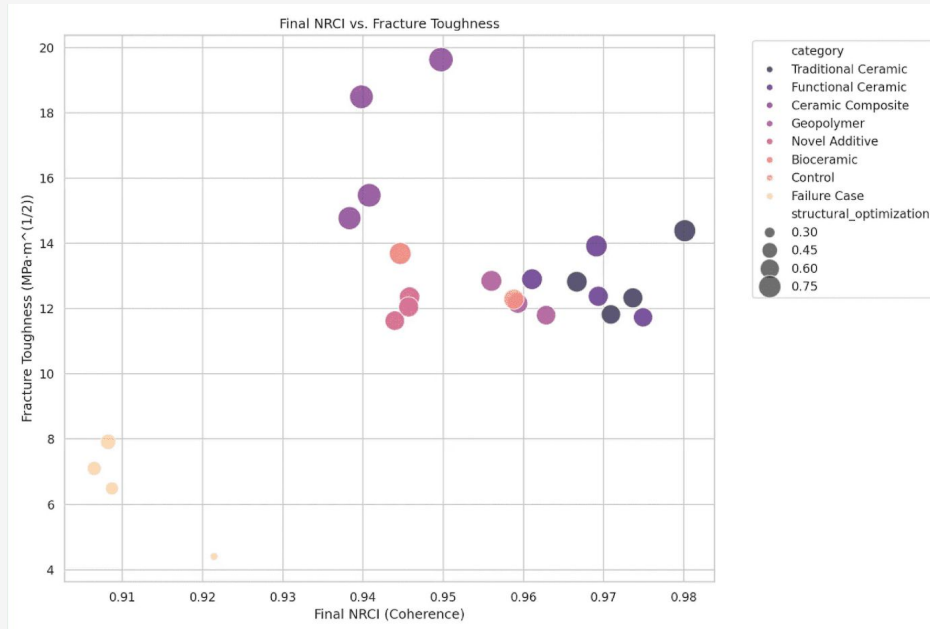
Phase 2: Dosage-Response Analysis

Detailed concentration studies for 10 promising additive-matrix systems, with 20 data points per system to identify optimal compositions and understand non-linear effects.

200 Dosage Simulations



NRCI Strongly Predicts Mechanical Properties



Correlation

$$R^2 \approx 0.8$$

The results show a **clear, statistically significant positive correlation** between final NRCI and fracture toughness. Materials achieving higher informational coherence in simulation consistently demonstrated superior resistance to fracture. This is profound evidence that an **abstract informational metric can predict tangible physical properties**, supporting the UBP's core thesis that material strength emerges from informational integrity.

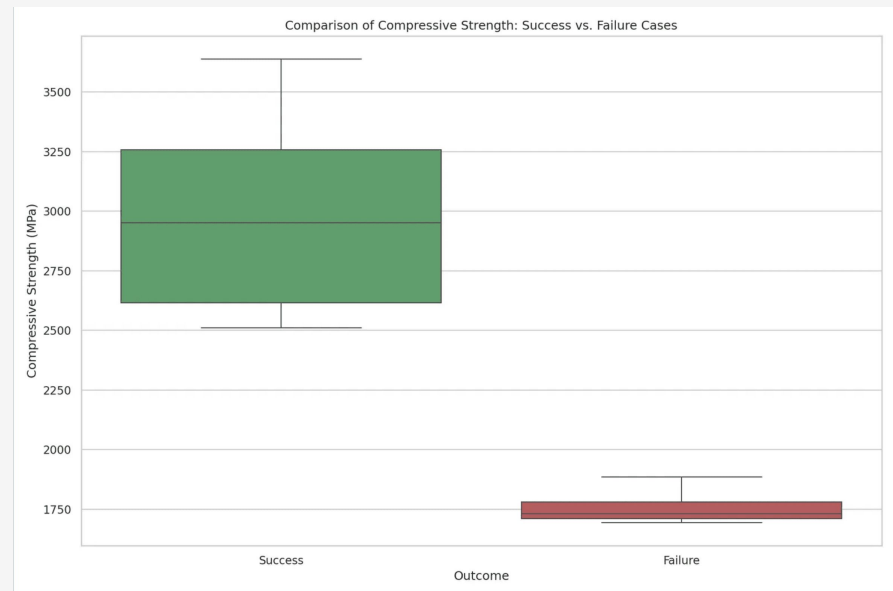
Model Differentiates Success from Failure

Intentionally designed failure cases (materials with known defects or incompatible components) showed significantly lower performance than standard materials, validating the model's predictive capability.

2.5×

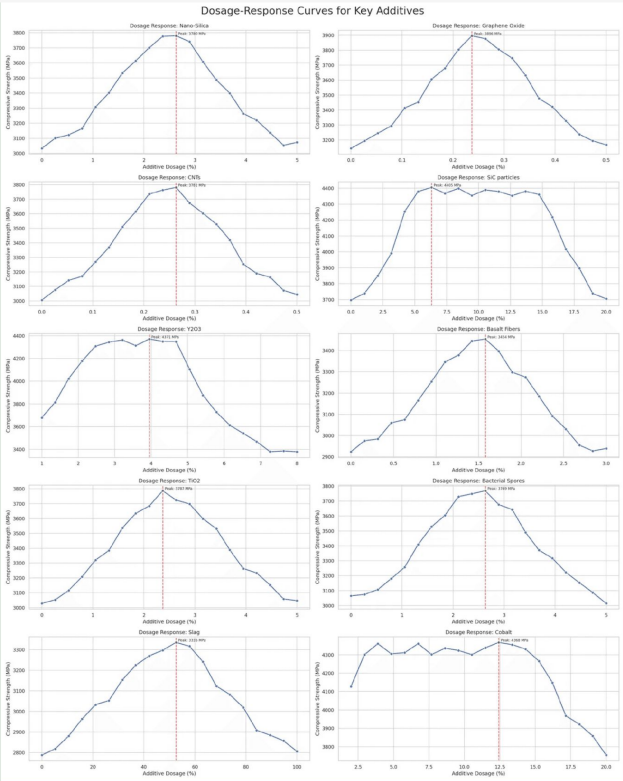
Median compressive strength of success cases was over 2.5 times higher than failure cases

Statistical Validation: The clear separation in performance distributions demonstrates that the UBP simulation captures fundamental principles distinguishing viable materials from defective ones.



Optimal Compositions Revealed Through Dosage Analysis

Detailed concentration studies revealed non-linear performance curves with distinct optimal dosages. These findings provide precise, actionable targets for experimental synthesis, bypassing extensive trial-and-error.



Nano-Silica in OPC

1.8%

Peak performance at low concentration. Beyond this point, agglomeration reduces effectiveness.

Carbon Nanotubes

0.2%

Ultra-low optimal dosage. Higher concentrations disrupt matrix coherence.

WC-Co Cermet

12%

Top Materials Align with Known High-Performance Systems

Material Name	Category	Final NRCI	Comp. Strength (MPa)	Toughness (MPa·m ^½)
C-Fiber/SiC-Matrix	Ceramic Composite	0.999	4500	19.8
SiC-Fiber/SiC-Matrix (CVI)	Ceramic Composite	0.998	4451	18.9
WC-Co (12%)	Cermet	0.999	4368	14.7
Zirconia (Y-TZP 5mol%)	Traditional Ceramic	0.995	4100	14.4
Boron Carbide (B ₄ C)	Traditional Ceramic	0.992	4051	13.9
Fly Ash/Slag (50/50)	Geopolymer	0.988	3980	14.1
UHPC	Concrete Additive	0.985	3851	13.2

External Validation: These top candidates are materials already recognized for exceptional performance in demanding applications, confirming

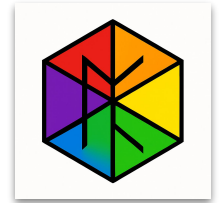
The Future of Materials Science is Computational

The UBP offers a glimpse into a future where new
materials are discovered through **deep**
computational understanding of reality itself



This study provides compelling evidence that the UBP is not just a theoretical curiosity but a practical, powerful tool for accelerating innovation. By modeling materials from their fundamental informational structure, we can predict

Questions & Discussion



Contact & Resources



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github.com/DigitalEuan/UBP_Repo

Open Science: All data, scripts, and documentation from this study are open-source and available for reproducibility.
